Jacob Democko

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Professional Summary: Senior Mechanical Engineer with a proven track record of managing high-tolerance mechanical components across complex supply chains. Combines technical expertise in precision design, DFM/DFA, and advanced manufacturing methods in pursuit of BOM optimization, cost reduction, and global supplier engagement. Adept at aligning engineering requirements with manufacturing capabilities to delivery highly reliable systems.

WORK EXPERIENCE

Lawrence Livermore National Lab **Aug. 2023 – Present**
Senior Mechanical Engineer *Livermore, CA*

- LLNL is a premier $3B+ federally funded research and development center (FFRDC) employing over 9,000 multidisciplinary experts dedicated to solving the nation's most complex nuclear security and energy challenges.
- As a senior mechanical engineer in Precision Systems & Manufacturing, I work on a variety of cutting-edge projects on advanced technical documentation, material selection, and design verification for precision mechanical assemblies, ensuring strict compliance with high-tolerance standards. Applied industry standards (ASME, ISO) to review and validate technical drawings, incorporating GD&T and tolerance analysis.
    - **Key Results:** Delivered a mirror-based metrology fixture and inspection process for OGP systems, enabling sub-10 μm measurement of previously inaccessible geometries and improving inspection throughput for high-precision components for fusion energy research at the National Ignition Facility.

Lucid Motors **May. 2021 – Aug. 2023**
Product Design Engineer *Newark, CA*

- Lucid Motors is a pioneering luxury electric vehicle startup with a $2.5B market cap, 6500 employees, and achieved production of the Air Sedan and Lucid Gravity SUV, delivering advanced automotive technology.
- As a production design engineer, I served as technical lead for core mechanical and thermal hardware components, driving supplier engagement and design teams, component qualification, and auditing supplier capabilities in casting, injection molding, precision machining and sheet metal stamping.
    - **Key Results:** Spearheaded a 50% unit cost reduction for front-end cooling module through design-for-manufacturability (DFM/DFA) initiatives, material substitutions, and optimizing component geometry for casting and molding processes. Conducted technical reviews and audits of tier-1 suppliers to identify production risks and resolve material defect or quality issues.

RMF Engineering **Jan. 2019 – Aug. 2020**
Mechanical Infrastructure Design Consultant *Raleigh, NC*

- Executed 20+ mechanical infrastructure projects from kickoff through construction administration and close-out

EDUCATION

University of California, Davis & Berkeley *Davis & Berkeley, CA*
M.S., Mechanical Engineering (Davis), Graduate Certificate in Technology Policy (Berkeley), Defense & Dual Use Technology Club
Coursework: Advanced Energy Systems, Micro & Nano Biotechnology, Sustainable Manufacturing, Technology & Nuclear Policy

- **GPA: 4.0/4.0**

North Carolina State University *Raleigh, NC*
B.S., Mechanical Engineering, graduated Summa Cum Laude

- **GPA: 3.89/4.0**

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** DOE Q Security Clearance; Engineer in Training (EIT)
- **Technologies:** CAD (Catia, Creo); CAE; Data Analysis (MATLAB, Python); KLayout; NI LabVIEW; Jira
- **Skills:** Communication; DFM/DFA; GD&T; Tolerance and Repeatability Analysis; Positive Attitude
- **Interests:** Fitness (Weightlifting, Running); Traveling; Restaurants, Automotive (Cars & Motorcycles); Comedy